

Dhvani Rakshak — Real-World Hackathon Judge Q&A;

16 Short & Impressive Answers Designed for Rapid Verbal Presentation

■ Category 1: Problem & Real-World Necessity

Q1: Why does India need Dhvani Rakshak right now when banks already use OTPs?

Ans: OTPs verify *what you have*, not *who is speaking*. Scammers use realistic AI voice clones over live calls to trick victims into sharing OTPs or approving wire transfers voluntarily.

Q2: Who suffers the most from AI voice clone scams in the real world?

Ans: Bank account holders, elderly citizens, corporate finance managers, and IAS/IPS VIPs targeted for deepfake extortion and wire fraud.

Q3: What is the financial scale of this problem in Indian banking?

Ans: Voice scam losses run into hundreds of crores annually, with single executive impersonation frauds exceeding **■50 Lakhs to ■2 Crores**.

Q4: How does Dhvani Rakshak stop fraud before money leaves the bank?

Ans: By analyzing live call audio in **under 200 ms** and automatically cutting the line or triggering OTP step-up verification before the transaction is authorized.

■ Category 2: Practical Deployment & Telecom Reality

Q5: How will a bank actually install Dhvani Rakshak into their call centers?

Ans: As a lightweight **SIP/RTP Media Gateway sidecar** that taps into existing PBX switches (Cisco/Avaya) via WebSockets without replacing legacy systems.

Q6: How do you handle poor 2G/4G cellular network coverage or background noise?

Ans: We normalize speech to **16kHz mono** and track fundamental pitch phase consistency (F0), which stays stable even through lossy mobile network compression.

Q7: What happens when 100,000 customers call a bank simultaneously during peak hours?

Ans: Our backend uses **Docker microservices**, Redis Pub/Sub async audio queues, and GPU model acceleration to handle thousands of concurrent calls seamlessly.

Q8: How will you commercialize or deploy this project in the market?

Ans: As a **B2B SaaS Security API** for banks, IVR providers, and government defense desks on a per-call subscription model.

■ Category 3: User Experience, False Positives & Edge Cases

Q9: What if a legitimate customer has a severe cold or sore throat? Won't your system block them?

Ans: No. A sore throat alters sound timbre but **preserves natural human pitch vibrations**. Borderline scores trigger an OTP check instead of a hard block.

Q10: How does a non-technical bank agent understand why a call was flagged as fake?

Ans: Through **Explainable AI (XAI)** cards that display plain-English reasons like '*Unnatural flat pitch detected*' or '*Zero human breathing*'.

Q11: Can scammers bypass your system by adding background traffic or rain noise?

Ans: No. We apply **spectral subtraction noise filtering** to isolate human vocal tract acoustics from environmental noise.

Q12: How fast does a new user need to speak to register their voice?

Ans: Just **10 seconds** of sample speech in our Enrollment Studio to generate a permanent mathematical voice fingerprint.

■ Category 4: Privacy, Legal Compliance & Uniqueness

Q13: If you listen to live banking calls, how do you comply with India's DPDP Act 2023 privacy law?

Ans: **Zero Raw Audio Retention**. Audio is processed in volatile RAM memory and immediately deleted after extracting numerical features.

Q14: How does Dhvani Rakshak store voice profiles without violating privacy?

Ans: Voices are converted into an **irreversible 192-dimensional numerical vector** (d-vector) encrypted with **AES-256**. Original speech cannot be reconstructed.

Q15: Why do you use Blockchain, and how does it help in court?

Ans: We stamp a **SHA-256 Merkle hash on Polygon Blockchain** to provide tamper-proof legal court evidence if a fraud dispute occurs.

Q16: What makes Dhvani Rakshak unique compared to existing products like Nuance?

Ans: Legacy tools only verify static voice prints; Dhvani Rakshak provides **real-time generative deepfake detection + XAI reasoning + Blockchain court proof**.